Predictive Maintenance Scheduling for Aircraft Engines Based on Remaining Useful Life Prediction

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***Abstract*—This article presents a novel data-driven predictive maintenance scheduling framework for aircraft engines based on remaining useful life (RUL) prediction. First, a deep learning ensemble model is proposed to effectively predict aircraft engine RUL, including a 1-D convolutional neural network (CNN) and a bidirectional long short-term memory network with an attention mechanism (Bi-LSTM-AM). Second, we propose a Bayesian optimization method to optimize the hyperparameters in the deep learning ensemble model to further improve RUL prediction performance. As the aircraft engine RUL decreases over time and eventually triggers a maintenance alarm threshold. The maintenance scheduling task is initiated after the aircraft engine maintenance alert threshold has been triggered. To effectively implement the maintenance scheduling plan, we develop a novel and effective mixed-integer linear programming (MILP) model to cope with aircraft engine maintenance scheduling, which aims to minimize the maximum maintenance time. Finally, experimental results show that our proposed data-driven predictive maintenance scheduling framework can monitor the running status of aircraft engines in real time and reduce their maintenance time.**

***Index Terms*—Bayesian optimization (BO), bidirectional long short-term memory (Bi-LSTM), convolutional neural network (CNN), maintenance scheduling, predictive maintenance, remaining useful life (RUL).**

I. Introduction

FOR MAINTENANCE problems of large equipment or systems, such as aircraft, maintenance costs represent a large percentage of total operating costs [1]. According to the International Air Transport Association (IATA), aircraft maintenance costs are estimated to account for 10.5% of an airline's operating expenses, and will cost 3.67 million dollars per aircraft in 2020 [2]. Therefore, reducing maintenance costs is of great importance. With the development of big data methods and Internet of Things (IoT) technologies, several effective predictive maintenance decision-making methods have been developed [3], [4], which provide great help in the maintenance of large equipment or systems. Currently, predictive maintenance in IoT focuses on monitoring the health of the system based on the information collected by the sensors [5]. Predictive maintenance aims to optimize maintenance decisions based on industrial systems' remaining useful life (RUL) predictions. How accurate RUL prediction is one of the core tasks of predictive maintenance as it can effectively help reduce the cost of maintenance. Developed sensing and IoT technologies have created a prerequisite for solving this problem by using data collected by multiple sensors at the same time as system degradation characteristics or health indicators. However, RUL estimation for industrial systems remains challenging due to the complexity of the operating environment. Therefore, there is a necessity to develop effective methods to predict the RUL of industrial systems to support predictive maintenance.

Existing predictive maintenance methods can be broadly categorized into two groups: 1) model-based methods and 2) data-driven predictive maintenance. Model-based predictive maintenance decisions mainly rely on stochastic processes (e.g., Gaussian processes, Markov processes, Poisson processes, and Weibull distributions [6], [7]) to characterize the degradation process of a system. Since stochastic processes are based on probability and statistical theories, and the model parameters are closely related to the degradation and failure processes of the system, predictive maintenance modeling based on stochastic processes has been widely studied [8], [9]. Predictive maintenance models are based on the assumption that the degradation mechanism of the system is consistent with the stochastic process [10], [11]. Therefore, the performance of predictive maintenance models based on stochastic processes relies on the quality of the a priori knowledge of the system's degradation process. However, in practice, systems typically face complex operational environments with degradation mechanisms that rarely follow this generalized degradation process. Therefore, even if a generalized degradation model is developed, it cannot be directly applied to modern industrial systems where a variety of operational variables exist [12]. Advances in IoT and sensor technologies have provided favorable help in solving this problem and applying data-driven methods to learn the degradation patterns of the system, which overcomes the drawbacks of model-based methods.

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In data-driven predictive maintenance decisions, machine learning or deep learning methods are used to predict the health state and RUL of the system operation without requiring a priori knowledge of the system degradation process [13], [14]. Compared with traditional maintenance decision-making methods, RUL prediction methods have been widely used in research, demonstrating their great potential for improving operational efficiency and reducing maintenance costs. For instance, Babu et al. [15] proposed a convolutional neural network (CNN) method to predict aircraft engine RUL, and the experimental results on the data set showed that CNN has superior RUL prediction performance compared to some machine learning methods. Zhang et al. [16] designed a multiobjective deep belief networks ensemble (MODBNE) approach to predict the aircraft engine RUL. They applied a multiobjective evolutionary algorithm to train both accuracy and multiobjective and combined the evolved DBN to form an ensemble model for the final RUL prediction. Li et al. [17] adopted a deep CNN (DCNN) method for RUL prognostics. Later, some researchers proposed a long short-term memory network (LSTM) model and its variants to predict RUL in industrial systems [18], [19], [20]. Based on LSTM, scholars further proposed a bidirectional LSTM network (Bi-LSTM) model and its variants for evaluating system RUL [21], [22]. In addition, Kim and Kaibo [23] designed a new Bayesian deep learning (BDL) framework for predicting interval estimates of RUL. Liu et al. [24] proposed a feature-attention-based bidirectional gated recurrent units (BGRUs) and CNN model (AGCNN) to predict system RUL. Zhang et al. [25] developed a novel BGRU with temporal self-attention mechanism (BiGRU-TSAM) to predict RUL. Recently, Zhang et al. [26] proposed a novel integrated multihead dual sparse self-attention network (IMDSSN) based on a modified Transformer to predict the RUL. Tian et al. [27] developed a LSTM framework based on spatial correlation and temporal attention mechanism (SCTA-LSTM) for evaluating system RUL.

Based on the above literature, it can be seen that most of the existing studies have considered the prediction of aircraft engine RUL, but have not considered the later maintenance decision-making for aircraft engines. To the best of our knowledge, studies integrating prediction results and predictive maintenance decisions for aircraft engines are limited. For example, Nguyen and Medjaher [12] proposed a dynamic predictive maintenance model based on predictive information of aircraft engines, which can efficiently estimate the probability of failure for aircraft engines in different periods, and then make maintenance decisions by quickly evaluating the required maintenance cost in each period, which aims to achieve effective maintenance of aircraft engines and timely ordering of spare part. Subsequently, Chen et al. [28] developed a deep auto-encoder and Bi-LSTM ensemble method that can effectively predict the health state and RUL of aircraft engines and make predictive maintenance decisions based on the prediction results. Lu et al. [29] proposed a data-driven dynamic predictive maintenance strategy and predicted component degradation based on online degradation data and continuously modified the maintenance schedule based on the prediction results, aiming to minimize the maintenance cost. Lee and Mitici [30] used reinforcement learning techniques to estimate the RUL distribution of aircraft engines and then optimized maintenance decisions based on the estimated RUL distribution. Shoorkand et al. [31] formulated a hybrid CNN-LSTM model to predict the aircraft engine RUL, and developed a framework for jointly optimizing production and imperfectly predicting maintenance schedules, which can effectively reduce the total production and maintenance costs. Recently, Wang et al. [32] proposed a CNN-Bi-LSTM ensemble method for predicting aircraft engine RUL and developed a dynamic predictive maintenance model based on the RUL prediction results. This dynamic predictive maintenance strategy includes order, stock, and maintenance decisions. Meanwhile, the influence of aircraft engine operating conditions on the predictive maintenance model is considered.

Although existing studies have considered predictive maintenance decision-making methods based on aircraft engine RUL predictions in their models, they have not systematically considered multiple aircraft engines for maintenance problems. Therefore, to fill the gap in the research area, we propose a novel data-driven predictive maintenance framework for aircraft engine maintenance. First, we propose a 1-D CNN and Bi-LSTM with an attention mechanism (CNN-Bi-LSTM-AM) ensemble model to predict the aircraft engine RUL through the sensor-collected time series data. To further improve RUL prediction performance, we propose a Bayesian optimization (BO) to optimize the hyperparameters in the CNN-Bi-LSTM-AM ensemble model, denoted as BO-CNN-Bi-LSTM-AM. The proposed BO-CNN-Bi-LSTM-AM ensemble model makes the prediction more accurate, thus enhancing the performance of predictive maintenance decision-making. Then, we consider a maintenance scheduling optimization problem for multiple aircraft engines when a maintenance alert threshold is triggered, which aims to minimize the maintenance time and thus reduce the idle time of the aircraft. Meanwhile, we formulate the maintenance scheduling problem for aircraft engines as a mixed-integer linear programming (MILP) model so that it can be solved exactly in a short period. The main contributions of this study compared to previous studies are as follows.

1) We present a novel data-driven predictive maintenance scheduling framework for aircraft engines based on RUL prediction.
2) A powerful BO-CNN-Bi-LSTM-AM ensemble model is developed to achieve effective aircraft engine RUL prediction.
3) We simultaneously consider the maintenance scheduling optimization problem for multiple aircraft engines after a maintenance alarm threshold is triggered and formulate it as a MILP model, which aims to minimize the maintenance time.

The remainder of this study is structured as follows. Section II describes in detail a complete BO-CNN-Bi-LSTM-AM ensemble model. In Section III, we describe the aircraft engine maintenance scheduling and formulate it as a MILP model. An aircraft engine data set from NASA is analyzed

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in Section IV, and the experimental results illustrate the effectiveness of the proposed predictive maintenance scheduling framework. Section V gives conclusions and discusses future research.

II. Methodology

A. Convolutional Neural Network Model

Convolution, pooling, and full connection (FC) layers are the three key components of CNN architecture. The convolution layers extract features through the convolution kernel, including performing element-level multiplication and summation operations on features. Then, the pooling layer reduces the dimension of the feature while retaining the most salient information. Finally, the main function of FC layer is to integrate the features calculated by the convolution layer and the pooling layer into a single value. In this article, we consider multidimensional data samples from multiple sensor detections X as input to CNN model

$$X = \{x_1, x_2, \ldots, x_i\} \quad \forall i \in N \tag{1}$$

where X is the number of samples and x_i is a data set for system operation status monitoring. CNN can effectively extract representative features hidden in monitoring data, but it is difficult to learn information related to temporal features. The monitoring data of system degradation usually contain a set of time series data and are inevitably related to the state of system degradation. To predict the degradation state of the system more effectively, the deep representative features and time series features related to the degradation of the system should be considered. According to [33], LSTM and Bi-LSTM model can effectively identify time-related features and is widely used in the prediction of system RUL. Therefore, we further introduce Bi-LSTM based on CNN model.

B. Bidirectional Long Short-Term Memory Network Model

LSTM architecture consists mainly of input gates, forgetting gates, and output gates. These gates regulate the flow of information within the network and determine which parts of the input should be memorized, forgotten, or output. Thus, LSTM can effectively capture long-term dependencies in the input sequence and make more accurate predictions using time series data. Compared with a single LSTM, Bi-LSTM can learn both forward and backward correlation information, thus helping to better predict system RUL. Therefore, Bi-LSTM has received a lot of attention in the research of system RUL prediction [34], [35]. a Bi-LSTM unit contains two LSTMs, which are used to process the state data monitored by the system from two different directions (i.e., forward and backward). For the input data x_i containing temporal features are processed by CNN into x_t and then updated by forward and backward LSTM (i.e., Bi-LSTM) as follows:

$$\vec{o}_t = \mathrm{LSTM}\left(\vec{x}_t, \vec{h}_t\right) = \begin{cases} \vec{f}_t = \mathrm{sig}\left(\vec{W}_f \cdot \vec{x}_t + \vec{U}_f \cdot \vec{h}_{t-1} + \vec{b}_f\right) \\ \vec{i}_t = \mathrm{sig}\left(\vec{W}_i \cdot \vec{x}_t + \vec{U}_i \cdot \vec{h}_{t-1} + \vec{b}_i\right) \\ \vec{C}_t^* = \tanh\left(\vec{W}_c \cdot \vec{x}_t + \vec{U}_c \cdot \vec{h}_{t-1} + \vec{b}_c\right) \\ \vec{C}_t = \vec{f}_t * \vec{C}_{t-1} + \vec{i}_t * \vec{C}_t^* \\ \vec{o}_t = \mathrm{sig}\left(\vec{W}_o \cdot \vec{x}_t + \vec{U}_o \cdot \vec{h}_{t-1}, + \vec{b}_o\right) \\ \vec{h}_t = \vec{o}_t * \tanh\left(\vec{C}_t\right) \end{cases} \tag{2}$$

$$\overleftarrow{o}_t = \mathrm{LSTM}\left(\overleftarrow{x}_t, \overleftarrow{h}_t\right) = \begin{cases} \overleftarrow{f}_t = \mathrm{sig}\left(\overleftarrow{W}_f \cdot \overleftarrow{x}_t + \overleftarrow{U}_f \cdot \overleftarrow{h}_{t-1} + \overleftarrow{b}_f\right) \\ \overleftarrow{i}_t = \mathrm{sig}\left(\overleftarrow{W}_i \cdot \overleftarrow{x}_t + \overleftarrow{U}_i \cdot \overleftarrow{h}_{t-1} + \overrightarrow{b}_i\right) \\ \overleftarrow{C}_t^* = \tanh\left(\overleftarrow{W}_c \cdot \overleftarrow{x}_t + \overleftarrow{U}_c \cdot \overleftarrow{h}_{t-1} + \overleftarrow{b}_c\right) \\ \overleftarrow{C}_t = \overleftarrow{f}_t * \overleftarrow{C}_{t-1} + \overleftarrow{i}_t * \overleftarrow{C}_t^* \\ \overleftarrow{o}_t = \mathrm{sig}\left(\overleftarrow{W}_o \cdot \overleftarrow{x}_t + \overleftarrow{U}_o \cdot \overleftarrow{h}_{t-1} + \overleftarrow{b}_o\right) \\ \overleftarrow{h}_t = \overleftarrow{o}_t * \tanh\left(\overleftarrow{C}_t\right) \end{cases} \tag{3}$$

where "→" and "←" denote the forward and backward processes in Bi-LSTM, respectively; f_t, i_t, and C_t^*, and o_t and h_t indicate the computational processes of input, forgetting, and output gates, respectively; in addition, C_t^* is the candidate cell state and C_t is the current cell state; o_t is the output of Bi-LSTM; W_f, W_i, W_c, and W_o are the input weights related to current input x_t; U_f, U_i, U_c, and U_o are the recurrent weights related to previous recurrent input h_{t-1} or future recurrent input h_{t+1}; b_f, b_i, b_c, and b_o are the bias vectors associated with Bi-LSTM states; and "sig" (i.e., logistic sigmoid) and "tanh" (i.e., hyperbolic tangent) are nonlinear activation functions, and "$*$" denotes point multiplication of two vectors.

Ultimately, the complete Bi-LSTM provides cascading vectors for forward and backward process outputs as follows:

$$o_t = \vec{o}_t \oplus \overleftarrow{o}_t \tag{4}$$

where "$\oplus$" denotes the summation by elements for the forward and backward outputs.

C. Attention Mechanism

In the prediction process of the system RUL, an effective operation is to make CNN-Bi-LSTM training model focus on different regions of interest by assigning different weights to different features at different time steps. In this process, since no prior information is available, an attention mechanism is introduced to learn more important information about features and time steps in CNN-Bi-LSTM model. Assume that the learned features by CNN-Bi-LSTM model for one sample can be expressed as $\boldsymbol{x_t}(m)$, where t is the time steps and m is the number of features. For the convenience of representation, $\boldsymbol{x_t}(m)$ is simplified to $\boldsymbol{x_i}$. Then, $\boldsymbol{x_i}$ is used as input to establish the mapping relationship between different time steps and features through CNN-Bi-LSTM model, and the corresponding importance γ_i is as follows:

$$\gamma_i = \sigma(W_i \boldsymbol{x_i} + b_i) \tag{5}$$

where W_i and b_i are the corresponding weight matrix and the bias vector, respectively. After the importance γ_i of the

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input sample is obtained at each moment, it is necessary to normalize the importance γ_i through a *softmax* function, as shown in

$$\alpha_i = \text{softmax}(\gamma_i) = \frac{e^{\gamma_i}}{\sum_{j=1}^{t} e^{\gamma_j}} \tag{6}$$

where α_i is the attention weight. The attention weight α_i is multiplied by the corresponding sample data $\boldsymbol{x_i}$ to obtain the output f_i. The output of the entire attention mechanism layer is F, which consists of the set of f_i outputs from each sample in the attention mechanism layer. The specific calculation formula is as follows:

$$f_i = \alpha_i \boldsymbol{x_i} \tag{7}$$

$$F = \{f_1, f_2, \ldots, f_i\}. \tag{8}$$

D. Bayesian Optimization Hyperparameters

In deep learning models, hyperparameters are usually set manually or empirically before training. However, as the amount of data increases, the time required to train the CNN-Bi-LSTM-AM model increases. Manual tuning of hyperparameters will inevitably reduce the efficiency. Meanwhile, the obtained prediction results may not be optimal [36]. Therefore, to improve the prediction performance and generalization ability of the CNN-Bi-LSTM-AM model, a BO method is proposed to optimize the hyperparameters of the CNN-Bi-LSTM-AM model.

First, hyperparameter optimization methods include various methods, such as gridsearch, random search, and BO, which BO has been proven to be very effective for algorithmic hyperparameters optimization [36], [37]. BO consists of two main parts: 1) the Gaussian process model and 2) the acquisition function. The Gaussian process model can be expressed as follows:

$$f(a) \sim GP\big[m(a), k\big(a, a'\big)\big] \tag{9}$$

where $m(a)$ is the mean function of $f(a)$ and $k(a, a')$ is the covariance function. The Gaussian process model can be used to obtain posterior probability distributions from the available data observations, while the acquisition function uses an alternative model derived from the Gaussian process model to select the next point to be evaluated.

Second, acquisition functions mainly include expected improvement (EI), probability of improvement, and the upper confidence bound. In this study, EI is chosen as the acquisition function and its formula is as follows:

$$E[I(a)] = \begin{cases} \big(\mu(a) - f(a^+)\big)\Phi(Z) + \sigma(a)\,\phi(Z), & \text{if } \sigma(a) > 0 \\ 0, & \text{if } \sigma(a) = 0 \end{cases} \tag{10}$$

$$Z = \frac{\mu(a) - f\big(a^+\big)}{\sigma(a)} \tag{11}$$

where $\mu(a)$ is the mean value obtained from the Gaussian process model at different probabilities, $f(a^+)$ indicates that as of the last iteration, the performance (prediction error) of the CNN-Bi-LSTN-AM model is optimal under the hyperparameter a^+, and $\Phi(\cdot)$, $\phi(\cdot)$, and $\sigma(\cdot)$ are the cumulative distribution function, probability density function, and standard deviation function associated with the Gaussian process model, respectively. The optimal solution for the EI criterion is $a^* = \arg\max_{a \in \mathcal{A}} E[I(a)]$, where a is the hyperparameter to be optimized and $\mathcal{A}$ represents the hyperparameter search space. The detailed procedure of BO hyperparameters is shown in Algorithm 1. According to Algorithm 1, we can obtain the optimal hyperparameter a^* in the CNN-Bi-LSTM-AM model.

Algorithm 1 BO Hyperparameters

Input: initialize hyperparameters a_0, search space $\mathcal{A}$, maximum number of iterations $\mathcal{N}$, Gaussian process model $f(a)$, and acquisition function $E[I(a)]$

Let $D_0 = \{a_0, f(a_0)\}$, $D_{t-1} = D_0$, $t = 0$.
for $(t < \mathcal{N})$ **do**
  According to the current Gaussian process model $f(a)$, maximize the acquisition function $E[I(a)]$ under D_{t-1}, i.e., $E[I(a|D_{t-1})]$, and obtain the next most potential evaluation points a_t, $a_t = \text{argmax } E[I(a|D_{t-1})]$.
  Compute the function value $f(a_t)$ under a_t; add $\{a_t, f(a_t)\}$ into D_t, $D_t = D_{t-1} \cup \{a_t, f(a_t)\}$.
  $t = t + 1$.
end for

Output: the optimal hyperparameter a^*

E. BO-CNN-Bi-LSTM-AM Ensemble Model

In this study, a BO-CNN-Bi-LSTM-AM ensemble model is proposed to predict the aircraft engine RUL using time-series sensor data. The ability of the ensemble model to combine multiple methods allows more complex patterns and relationships to be captured in the data. First, CNN model can extract local features from high-dimensional inputs and transfer them to lower dimensional features [31]. Second, Bi-LSTM model can remember and connect previous information into the currently acquired data, capturing remote dependencies through several gates, thus improving the performance of the model [35]. Then, the attention mechanism further extracts the important information between the time window steps and features. Finally, BO hyperparameters are used to further improve the prediction performance of this ensemble model. Fig. 1 illustrate the overall structure of BO-CNN-BLSTM-AM ensemble model.

III. Predictive Maintenance Scheduling Based on Predicted RUL

A. Problem Description

Based on the predicted RUL, we can perform predictive maintenance on aircraft engines in real time. Given a set of aircraft engines $V = \{1, 2, \ldots, n\}$ that need to be maintained and a set of maintenance centers $M = \{1, 2, \ldots, m\}$. Each aircraft engine i has a maintenance time p_i^m associated with a maintenance center and a arrival time t_i^m. The arrival time t_i^m depends on the distance between the aircraft engine i and the maintenance center m. In the process of maintenance scheduling for aircraft engines, how to allocate aircraft engines to maintenance centers to minimize maintenance time is a key issue that needs to be optimized. Based on the number of aircraft engines to be maintained, we need to determine at which maintenance center the aircraft engines will be maintained to minimize the maintenance time of the aircraft

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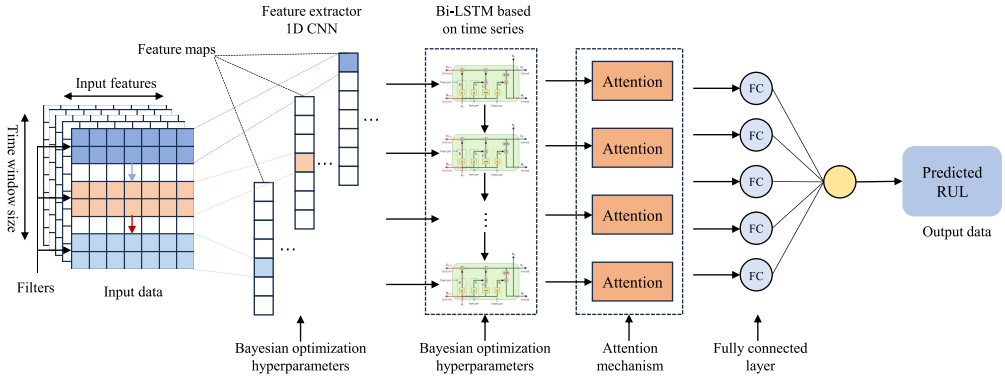

Fig. 1. Structure of BO-CNN-Bi-LSTM-AM ensemble model for RUL prediction.

engines. Thus, we propose a MILP model to deal with the aircraft engine maintenance scheduling problem. Before developing the optimization model, we make the following assumptions: 1) only one aircraft is allowed to be maintained in each maintenance center; 2) at most one aircraft engine for each maintenance center is under maintenance simultaneously; and 3) the number of maintenance centers is smaller than the number of aircraft due to the huge costs involved in constructing maintenance centers.

B. Notations

Indices:

i, j: Node index.

m: Maintenance center index.

Parameters:

V: Set of aircraft engines to be maintained, $V = \{1, 2, \ldots, n\}$.

M: Set of maintenance centers, $M = \{1, 2, \ldots, m\}$.

$|M|$: Number of maintenance centers.

t_i^m: Time required for aircraft engine i to reach the maintenance center m.

p_i^m: Maintenance time for aircraft engine i in maintenance center m.

L_i: Priority level of aircraft engine i, which indicates a smaller RUL for aircraft engine i.

t_0: Initial maintenance time of the aircraft engine at the maintenance centers.

Q: Large positive number.

Decision Variables:

x_{ij}: 1 if aircraft engine j is an immediate successor of aircraft engine i in the same maintenance center; 0 otherwise.

y_i^m: 1 if aircraft engine i is the first aircraft engine processed in the maintenance center m; 0 otherwise.

z_i^m: 1 if aircraft engine i is assigned to maintenance center m for processing; 0 otherwise.

C_i: Maintenance completion time of aircraft engine i.

C_{max}: Total maintenance completion time of all aircraft engines.

C. Maintenance Scheduling for Aircraft Engine

In the data-driven predictive maintenance framework, the aircraft engine system monitoring point makes the basic decision of whether to maintain or not based on the aircraft engine RUL. In addition, maintenance operations are considered to be perfect operations by replacing aircraft engines, so when is the best time to replace them? Meanwhile, aircraft engines cannot be replaced when the aircraft is on a mission. Therefore, we assume that maintenance scheduling is performed only when the aircraft is idle.

If the aircraft engine system is still operating during the running cycle, a maintenance decision is considered to be executed or not based on predicted RUL from the available monitoring data. Since maintenance cannot be performed during mission execution, it is necessary to know if the current RUL is lower than the mission cycle. If the answer is "yes," this implies that predictive maintenance must be performed because the aircraft engine system may fail while performing its mission. Thus, the decision for the predictive maintenance of aircraft engine i is made by

$$\text{Maintenance} = \begin{cases} 1, & \widehat{\text{RUL}}_i \leq \Delta t \\ 0, & \text{others} \end{cases} \tag{12}$$

where Maintenance $= 1$ denotes that predictive replacement is required to ensure mission execution, Maintenance $= 0$ denotes that predictive replacement is not required, $\widehat{\text{RUL}}_i$ denotes the predicted RUL of aircraft engine i, and Δt is a maintenance alarm threshold, which is determined according to the mission cycle of the aircraft engine.

If the aircraft engine RUL is less than the maintenance alarm threshold, then the aircraft is to be maintained, otherwise, a failure may occur. Given a maintenance alarm threshold Δt, when the number of n aircraft engines RUL is less than the threshold Δt, then these aircraft engines need to be maintained simultaneously. To effectively maintain aircraft engines, we mainly consider the following four types of constraints to formulate a maintenance scheduling model for the aircraft engine.

1) Aircraft Engine Assignment Constraints: The aircraft engine assignment constraint restricts each aircraft engine i to be assigned to a maintenance center at most once. We adopt a binary variable z_i^m to indicate whether aircraft engine i is assigned to maintenance center m for maintaining and then y_i^m to indicate that the first aircraft engine i to be maintained at the maintenance center m. The aircraft engine assignment

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constraints can be formulated as follows:

$$\sum_{m\in M} z_i^m = 1 \quad \forall i \in V \tag{13}$$

$$\sum_{i\in V} y_i^m = 1 \quad \forall m \in M \tag{14}$$

$$y_i^m \le z_i^m \quad \forall m \in M, i \in V. \tag{15}$$

Constraint (13) ensures that each aircraft engine is assigned to at most one maintenance center for maintenance. Constraint (14) implies that for each maintenance center, there exists the first aircraft engine to be maintained, thus ensuring that the maintenance center maintains the aircraft engine. Constraint (15) indicates that if an aircraft engine is not maintained at a maintenance center, it cannot be the first maintenance aircraft engine at the maintenance center.

2) Aircraft Engine Sequence Constraints: The aircraft engine sequence constraint ensures that all assigned aircraft engines can be maintained in sequence. Therefore, we propose a binary variable x_{ij} to constrain the maintenance sequence of aircraft engines i and j as follows:

$$\sum_{m\in M} y_j^m + \sum_{i\in V, i\neq j} x_{ij} = 1 \quad \forall j \in V \tag{16}$$

$$\sum_{i\in V, i\neq j} x_{ij} \le 1 \quad \forall j \in V \tag{17}$$

$$x_{ij} + z_i^m \le 1 + z_j^m \quad \forall m \in M, j, i \in V, i \neq j \tag{18}$$

$$x_{ij} + z_j^m \le 1 + z_i^m \quad \forall m \in M, j, i \in V, i \neq j. \tag{19}$$

Constraint (16) indicates that if aircraft engine j is the first aircraft engine maintained by the maintenance center, it cannot maintain any aircraft engine before it. Otherwise, it will have an aircraft engine that is previously under maintenance. Constraint (17) is a valid inequality indicating that each aircraft engine j is preceded by at most one aircraft engine under maintenance. Constraints (18) and (19) associate the assignment variable z_i^m to the sequence variable x_{ij}, which ensures that two consecutively maintained aircraft engines must be assigned to the same maintenance center. In particular, (18) indicates that if $x_{ij} = 1$ and aircraft engine i is assigned to maintenance center m, then it is ensured that aircraft engine j is maintained at maintenance center m. Constraint (19) indicates that if aircraft engines i and j are consecutively maintained aircraft engines at the same maintenance center m and maintenance center m maintains aircraft engine j, then aircraft engine i must be assigned to it for maintenance.

3) Maintenance Priority Constraints: In the process of maintaining aircraft engines, we also need to consider the maintenance time window. In addition, aircraft engines RUL may be different and there exists a maintenance time window for each aircraft engine. For convenience, we consider that the smaller an aircraft engine RUL is, the more urgent the maintenance window is, i.e., it needs to be prioritized to be maintained to fully utilize other aircraft-related facility resources. To achieve this, we introduce maintenance prioritization to cope with the maintenance time window of aircraft engines, i.e., different maintenance priorities depending on the size of the aircraft engine RUL. Without loss of generality, we adopt L_i to denote the maintenance priority of the aircraft engine, i.e., the smaller the aircraft engine $\widehat{\mathrm{RUL}}_i$, the higher the maintenance priority L_i. For simplicity, we first sort all aircraft engines $1, 2, \ldots, n$ in nonincreasing order of their maintenance priority. In addition, all maintenance centers $|M|$ should give priority to maintaining the first $|M|$ aircraft engines, i.e., aircraft engines $1, 2, \ldots, |M|$. The remaining aircraft engines shall be maintained based on their maintenance priorities

$$\sum_{m\in M}\sum_{i=1}^{|M|} y_i^m = |M| \tag{20}$$

$$x_{ij} = 0 \quad \forall j, i \in V, \text{if } L_i \ge L_j, \widehat{\mathrm{RUL}}_i \le \widehat{\mathrm{RUL}}_j. \tag{21}$$

Constraint (20) indicates that the first $|M|$ aircraft engines with higher maintenance priority are preferentially maintained by maintenance centers. Constraint (21) is a maintenance priority constraint for aircraft engines indicating that a low priority will not be maintained before a high-priority aircraft engine. Constraints (20) and (21) together ensure the maintenance priority of the aircraft engine.

4) Maintenance Completion Time Constraints: Maintenance completion time constraints ensure that aircraft engines can be maintained in a timely manner, reducing maintenance time and fully utilizing other aircraft-related facility resources. In addition, the maintenance completion time constraint describes the interrelationship of the maintenance completion time for each aircraft engine, which can be formulated as follows:

$$C_0 = t_0 \tag{22}$$

$$C_i \ge \sum_{m\in M} z_i^m \left(p_i^m + t_i^m\right) + C_0 \quad \forall i \in V \tag{23}$$

$$C_j \ge C_i + \sum_{m\in M} z_j^m p_j^m + Q\left(x_{ij} - 1\right) \quad \forall j, i \in V, i \neq j$$
$$\text{if } C_i \ge \widehat{\mathrm{RUL}}_j + t_j^m \tag{24}$$

$$C_j \ge \widehat{\mathrm{RUL}}_j + t_j^m + \sum_{m\in M} z_j^m p_j^m + Q\left(x_{ij} - 1\right) \quad \forall j, i \in V$$
$$i \neq j, \text{if } C_i \le \widehat{\mathrm{RUL}}_j + t_j^m \tag{25}$$

$$C_{\max} \ge C_j \quad \forall j \in V. \tag{26}$$

Constraint (22) denotes the initial maintenance time of the aircraft engine at the maintenance center, where 0 is a virtual aircraft engine to record the initial maintenance time. Similar formulation techniques can also be found in [38], [39], and [40]. Constraint (23) ensures that the maintenance completion time of each aircraft engine cannot be less than its maintenance time and arrival time at the maintenance center plus the initial maintenance time. Constraint (24) implies that if two aircraft engines are serviced consecutively at the same maintenance center, the maintenance completion time of the latter is not less than its maintenance time plus the completion time of the former when the maintenance time of the former is greater than $\widehat{\mathrm{RUL}}_j + t_j^m$. Constraint (25) shows that if two aircraft engines are maintained consecutively at the same maintenance center, the maintenance completion time of the latter is not less than its maintenance time plus the arrival time and the predicted $\widehat{\mathrm{RUL}}_j$ when the maintenance time of the former is less than $\widehat{\mathrm{RUL}}_j + t_j^m$. Clearly, (25) considers

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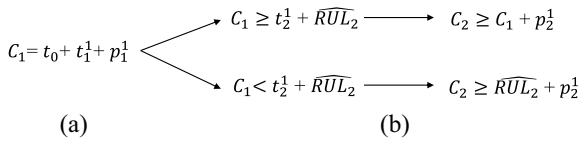

Fig. 2. Schematic for calculating aircraft engine maintenance time. (a) First aircraft engine. (b) Second aircraft engine.

the effect of the predicted RUL of the aircraft engine on the maintenance completion time. Constraint (26) ensures that the maintenance completion time for each aircraft engine is less than the maximum maintenance completion time.

In addition, we use Fig. 2 to further illustrate the calculation of maintenance completion time for aircraft engines at maintenance centers. In Fig. 2(a) the calculation of the maintenance completion time for the first aircraft engine arriving at the maintenance center 1 is represented. Then, the maintenance completion time of the first aircraft engine at maintenance center 1 is compared with the predicted RUL and arrival time at maintenance center 1 of a subsequent second aircraft engine. At this point, there are two possibilities: 1) is $C_1 \geq t_2^1 + \widehat{\mathrm{RUL}}_2$, then the maintenance completion time of the second aircraft engine is calculated by (24) and 2) is $C_1 < t_2^1 + \widehat{\mathrm{RUL}}_2$, then the maintenance completion time of the second aircraft engine is calculated by (25). Ultimately, the maintenance completion time of the second aircraft engine at maintenance center 1 can be calculated as shown in Fig. 2(b).

Finally, the decision variable constraints are defined as follows:

$$C_j \geq 0 \quad \forall j \in V \tag{27}$$

$$x_{ij} \in \{0, 1\} \quad \forall i, j \in V \tag{28}$$

$$y_i^m \in \{0, 1\} \quad \forall m \in M, i \in V \tag{29}$$

$$z_i^m \in \{0, 1\} \quad \forall m \in M, i \in V. \tag{30}$$

Based on the above process, a MILP model is formulated as follows:

$$\begin{aligned} &\min \ C_{\max} \\ &\text{s.t. } (13)\text{–}(30). \end{aligned} \tag{31}$$

Objective (31) represents to minimize the total maintenance completion time of all aircraft engines, thereby reducing the idle time of the aircraft at maintenance centers.

IV. Experimental Validation

A. Data Description

In this section, we use a simulation data set widely used for aircraft engine degradation performance testing to demonstrate how the proposed data-driven predictive maintenance scheduling framework can be applied in engineering. The data set is generated by the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) tool and simulates various degradation scenarios for a fleet of homogeneous type aircraft engines. The data set is divided into four subdata sets, FD001, FD002, FD003, and FD004, as shown in Table I. Meanwhile, each subdata set contains a training set and a test set. The training set contains all the data for all the samples from the initial state to the state in which the failure occurs, while the test set provides data only for the first part of the life cycle, which is intended for the subsequent RUL of the aircraft engine.

TABLE I
Details About C-MAPSS Data Set [41]

Datasets	FD001	FD002	FD003	FD004
Training instances	100	260	100	249
Test instances	100	259	100	248
Operating conditions	1	6	1	6
Fault modes	1	1	2	2
Training samples	20631	53759	24720	61249
Test samples	13096	33991	16596	41214

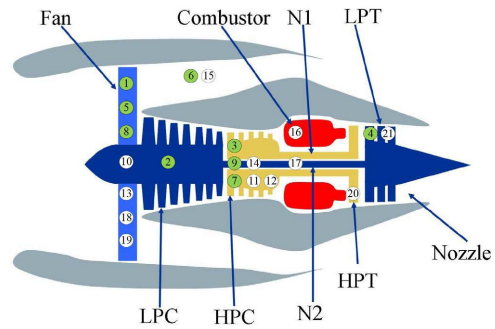

Fig. 3. Schematic of the structure and sensor location of an aircraft engine [41].

In addition, the data set consists of 26 columns that characterize the aircraft engine operations. The first and second columns represent the ID and degradation time step of each aircraft engine, respectively, the third through fifth columns describe the three modes of operation of the aircraft engines, and the last 21 columns correspond to the data monitored by the sensors. These sensors are deployed at different locations to monitor the operating conditions (e.g., speed, temperature, and pressure) of the aircraft engine, as shown in Fig. 3. Fig. 3 illustrates the structure of the main components of the aircraft engine, including the fan, low-pressure aircraft (LPT), low-pressure compressor (LPC), high-pressure compressor (HPC), high-pressure aircraft (HPT), etc. The corresponding descriptions of these sensors are provided in Table II. The sensors with IDs 1, 5, 6, 10, 16, 16, 18, and 19 in Table II always have constant values during the degradation of the aircraft engine lifetime, which means that the data obtained from these sensors have little influence with the degraded state of the aircraft engine. Therefore, these sensor data were removed from the training and test data sets. Finally, 14 sensors are used for aircraft engine RUL prediction.

In the experiments, the relevant parameters are set as shown in Table III. As shown in Table III, the hyperparameters to be optimized mainly include the number of neurons in CNN layer, the number of neurons in Bi-LSTM layer, dropout rate, activation function, and learning rate. Meanwhile, the time window step in the data set is set to 30. RUL prediction is a regression problem, thus the mean square error (MSE) is used as the loss function. Adam algorithm is utilized to train the

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TABLE II
DESCRIPTION OF THE 21 SENSOR DATA [41]

ID	Symbol	Description	Units
1	T2	Total temperature at fan inlet	°R
2	T24	Total temperature at LPC outlet	°R
3	T30	Total temperature at HPC outlet	°R
4	T50	Total temperature at LPT outlet	°R
5	P2	Pressure at fan inlet	psia
6	P15	Total pressure in bypass-duct	psia
7	P30	Total pressure at HPC outlet	psia
8	Nf	Physical fan speed	rpm
9	Nc	Physical core speed	rpm
10	epr	Engine pressure ratio (P50/P2)	–
11	Ps30	Static pressure at HPC outlet	psia
12	phi	Ratio of fuel flow to Ps30	pps/psi
13	NRf	Corrected fan speed	rpm
14	NRc	Corrected core speed	rpm
15	BPR	Bypass ratio	–
16	farB	Burner fuel-air ratio	–
17	htBleed	Bleed enthalpy	–
18	Nf_dmd	Demanded fan speed	rpm
19	PCNfR_dmd	Demanded corrected fan speed	rpm
20	W31	HPT coolant bleed	lbm/s
21	W32	LPT coolant bleed	lbm/s

TABLE III
CONFIGURATIONS OF THE HYPERPARAMETERS RANGE IN THE BO-CNN-BI-LSTM-AM ENSEMBLE MODEL

Hyperparameter	Range
CNN layers	(32, 128)
Bi-LSTM layers	(32, 128)
Dropout rate	(0.1, 0.4)
Activation function	(ReLu, Sigmoid)
Learning rate	(0.01, 0.0001)
Optimizer	Adam
Time window step	30
Batch size	64
Epoch	32
Loss function	Mean square error

TABLE IV
PARAMETERS CONFIGURATION OF BO-CNN-BI-LSTM-AM ENSEMBLE MODEL

Hyperparameter	FD001	FD002	FD003	FD004
CNN layers	118	64	111	88
Bi-LSTM layers	119	64	78	125
Dropout rate	0.26	0.30	0.23	0.29
Activation function	Sigmoid	ReLu	ReLu	ReLu
Learning rate	0.0039	0.0010	0.0015	0.0044

BO-CNN-Bi-LSTM-AM ensemble model. It is worth noting that changes in the data set or training set can lead to changes in the optimal hyperparameters. Then, BO is used to determine the optimal hyperparameters according to Table III. For the subdata sets FD001 to FD004, the optimized hyperparameters are shown in Table IV. Meanwhile, considering that the aircraft engine RUL is not significantly degraded during the initial state, we use the piecework linear function with elbow point 125 to correct the RUL label according to the data distribution of the aircraft engine RUL.

B. Discussion of Prognostics Performance

1) Evaluation Indexes of Model Performance: In the system RUL prediction model, the multisensor monitoring data reflecting the degradation characteristics of the aircraft engine are used as inputs, and the predicted RUL is utilized as the model output. In order to evaluate the effectiveness of BO-CNN-Bi-LSTM-AM ensemble model, two different metrics are considered: 1) root MSE (RMSE) and 2) score function. For RMSE, the purpose is to measure the deviation between the true RUL and predicted RUL, which is calculated as follows:

$$\mathrm{RMSE}_{\mathrm{RUL}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(d_i)^2} \tag{32}$$

where N is the number of test samples and $d_i = \widehat{\mathrm{RUL}}_i - \mathrm{RUL}_i$ is the difference between predicted and true RUL for aircraft engine i. Unlike RMSE, the score function takes into account the different penalties on the RUL predictions and is calculated as shown in (33). For example, RUL is penalized for being underestimated or overestimated (i.e., $d_i < 0$ or $d_i \geq 0$), and the penalty is greater for being overestimated. This is well explained by the fact that the cost of unplanned downtime is much higher than the cost of advance maintenance [18]

$$\mathrm{Score}_{\mathrm{RUL}} = \begin{cases} \sum_{i=1}^{N} e^{-\frac{d_i}{13}} - 1, & \text{if } d_i < 0 \\ \sum_{i=1}^{N} e^{\frac{d_i}{10}} - 1, & \text{if } d_i \geq 0. \end{cases} \tag{33}$$

2) Competitive Results Compared to Other Methods: To evaluate the performance of BO-CNN-Bi-LSTM-AM ensemble model, this article compares several state-of-the-art prognosis methods, as shown in Table V. From Table V, it can be seen that the proposed model outperforms existing study methods, such as CNN, LSTM, Bi-LSTM-ED, BDL, IMDSSN, and SCTA-LSTM, which indicates the powerful learning capability of the proposed model. Meanwhile, for the subdata sets FD001, FD002, and FD003, the proposed model outperforms the existing methods in both RMSE and score. Specifically, the proposed model is superior compared to [27] method on the four subdata sets from FD001 to FD004, the RMSE is improved by $(12.10 - 11.43)/11.43 * 100\% = 5.86\%$, $(16.90 - 15.69)/15.69 * 100\% = 7.71\%$, $(12.14 - 11.28)/11.28 * 100\% = 7.62\%$, and $(21.93 - 18.35)/18.35 * 100\% = 19.51\%$, respectively.

In addition, Fig. 4 demonstrates the prediction results of the proposed model for the four subdata sets RUL from FD001 to FD004. From Fig. 4, it can be seen that the proposed model predicts the RUL close to the actual RUL and can predict the degradation trend of the aircraft engine well, which demonstrates the effectiveness of the proposed model for RUL prediction. In particular, the accuracy of the model predictions tends to be higher in the region where the RUL values are small. This is because the failure characteristics are substantially enhanced when the aircraft engine operates close to failure and can be well captured by the proposed prediction model, resulting in better prediction results.

To further test the prediction performance of the proposed model, we also test a publicly available data set from the International Conference on prognostics and health management in 2008 (PHM08). The composition of this PHM08 data set is similar to the C-MAPSS data set, which is available for download on the NASA online website. The data set has 218 aircraft engines and includes both training and test sets.

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TABLE V
COMPARISON OF RUL PREDICTION PERFORMANCE

Method	FD001		FD002		FD003		FD004	
	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
MLP [16]	16.78	560.59	28.78	14026.72	18.47	479.85	30.96	10444.35
VLSTM [20]	15.23	250.00	22.87	4532.00	14.53	1523.00	26.11	5627.00
DBN [16]	15.21	417.59	27.12	9031.64	14.71	442.43	29.88	7954.51
MODBNE [16]	15.04	334.23	25.05	5585.34	12.51	421.91	28.66	6557.62
Bi-LSTM-ED [22]	14.74	273.00	22.07	3099.00	17.48	574.00	23.49	3202.00
Attention-LSTM [18]	14.53	322.44	-	-	-	-	27.06	5649.14
MCLSTM [19]	13.71	315.00	-	-	-	-	23.81	4826.00
Bi-LSTM [21]	13.65	295.00	23.18	4130.00	13.74	317.00	24.86	5430.00
LSTM [17]	13.52	431.70	24.42	14459.00	13.54	347.30	24.21	14322.00
DCNN [17]	12.61	273.70	22.36	10412.00	12.64	284.10	23.31	12466.00
BiGRU-TSAM [25]	12.56	213.35	18.94	2264.13	12.45	232.86	20.47	3610.34
AGCNN [24]	12.42	225.51	19.43	1492.00	13.39	227.09	21.50	3392.00
BDL [23]	12.19	267.21	18.49	2007.81	12.07	409.39	19.41	**2415.71**
IMDSSN [26]	12.14	206.11	17.40	1775.15	12.35	229.54	19.78	2852.81
SCTA-LSTM [27]	12.10	207.00	16.90	1267.00	12.14	248.00	21.93	3310.00
The proposed model	**11.43**	**201.26**	**15.69**	**1214.47**	**11.28**	**181.99**	**18.35**	2627.11

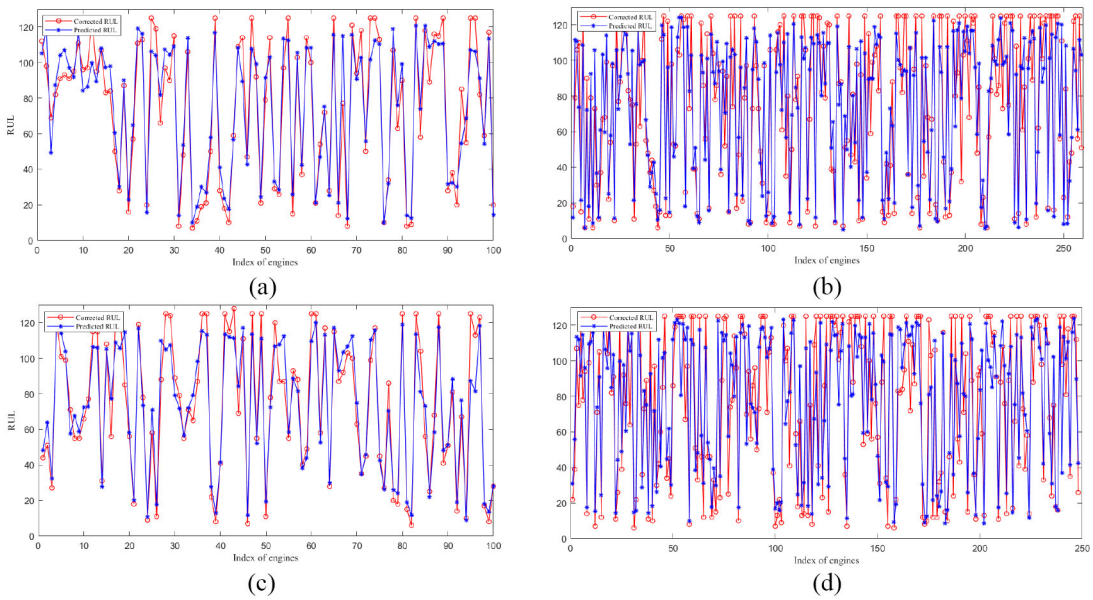

Fig. 4. RUL prediction results of the proposed model in the FD001 to FD004 test set. (a) RUL prediction results on FD001. (b) RUL prediction results on FD002. (c) RUL prediction results on FD003. (d) RUL prediction results on FD004.

However, the true RUL results for the test set are not available. Therefore, to evaluate the prediction results of this data set, we split the 218 aircraft engines in the training set into the training set and the test set by 8:2, where the first 174 aircraft engines are used as the training set and 175-218 aircraft engines are used as the test set to evaluate the performance of the proposed model. The results on the PHM08 data set are shown in Table VI. In Table VI, the hyperparameters are 87 for CNN layer, 112 for Bi-LSTM layer, a dropout rate of 0.23, an activation function of "Relu," and a learning rate of 0.0015 optimized by BO. From Table VI, it can be found that the proposed model has better prediction performance on RMSE compared to other deep learning models.

TABLE VI
EXPERIMENTAL RESULTS ON THE PHM08 DATA SET

Method	RMSE
CNN	56.52
LSTM	50.88
Bi-LSTM	48.73
CNN-LSTM	45.39
CNN-Bi-LSTM	43.45
The proposed model	**41.33**

C. Predictive Maintenance Scheduling

First, based on the prediction results in Table V and Fig. 4, we will use the test results in FD001 as a case study to verify the effectiveness of the proposed predictive maintenance scheduling model. Meanwhile, Fig. 5 corresponds to the number distribution of the predicted RUL interval for 100 aircraft engines in Fig. 4(a), whose purpose is to carry out predictive maintenance of aircraft engines. According to Fig. 5, it can be seen that when the maintenance alarm threshold $\Delta t = 10$, 6, 11, 20, and 18 aircraft engines need to be maintained in the RUL intervals 80 to 90, 90 to 100, 100 to 110, and 110

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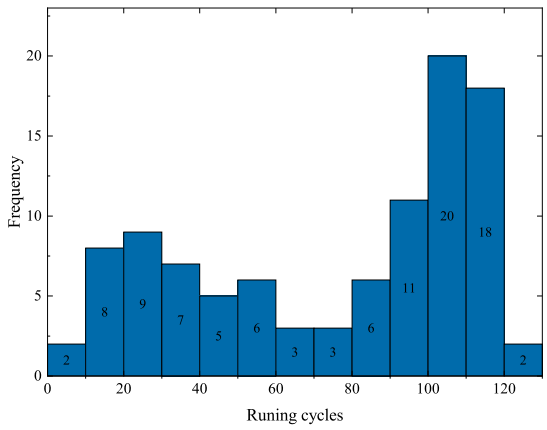

Fig. 5. RUL prediction intervals for 100 aircraft engines in the FD001 test set.

TABLE VII
RUL FOR AIRCRAFT ENGINE PREDICTION

Aircraft engines	1	2	3	4	5	6
$\widehat{RUL}$	81.82	84.43	86.54	87.26	89.32	89.58

TABLE VIII
MAINTENANCE TIME FOR AIRCRAFT ENGINES

Aircraft engines	1	2	3	4	5	6
Maintenance time	1.53	1.75	1.09	1.19	1.49	1.19

TABLE IX
ARRIVAL TIME OF AIRCRAFT ENGINES AT MAINTENANCE CENTERS

t_i^m	1	2	3	4	5	6
1	0.90	0.82	0.54	0.50	0.59	0.76
2	0.59	0.52	0.74	0.99	0.99	0.79
3	0.61	0.96	0.91	0.68	0.69	0.89

to 120, respectively. For these aircraft engines, it is necessary to consider where they are maintained and minimize their maintenance time to fully utilize the performance of other components of the aircraft. First, we give the predicted RULs for the 6 aircraft engines in the RUL intervals 80 to 90 as shown in Table VII. In order to facilitate solving the studied problem, we first preprocess the RULs of these 6 aircraft engines. Taking the first aircraft engine as an example, let its RUL be 0, then the second aircraft engine RUL is 2.61, the third aircraft engine RUL is 4.72, and so on, the 6 aircraft engine RUL is 7.76. In other words, we make C_0 is equal to 0 by translating the aircraft engine RUL. Meanwhile, the required maintenance time for these 6 aircraft engines is randomly generated in [1.00, 2.00] and the arrival time at the maintenance center is randomly generated in [0.50, 1.00]. We study the maintenance under 2 and 3 maintenance centers, whose corresponding data are shown in Tables VIII and IX.

Based on the above data, we adopt CPLEX 12.80 to solve the MILP model and the results are shown in Table X. In Table X, the first column M denotes the maintenance center, the second column *obj* denotes the objective function, the third column denotes the solution time of CPLEX, the fourth column denotes the optimal scheduling plan, MILP denotes the proposed model, and MILP-1 denotes the model

TABLE X
MAINTENANCE SCHEDULING RESULTS FOR SIX AIRCRAFT ENGINES

M	MILP			MILP-1		
	obj	CPU (s)	Plan	*obj*	CPU (s)	Plan
1	9.74	0.02	2-3-4-5	4.85	0.03	2-3-4
2			1-6			1-5-6
1			2-6			3-5
2	9.71	0.10	1-4	3.46	0.03	2-6
3			3-5			1-4

TABLE XI
RESULTS WITH DIFFERENT NUMBERS OF AIRCRAFT ENGINES

V	M	MILP		MILP-1	
		obj	CPU (s)	*obj*	CPU (s)
11	2	12.11	0.11	8.59	0.05
11	3	11.45	0.19	5.83	0.16
18	3	11.76	1.58	9.26	9.94
18	4	11.64	3.02	7.05	6.09
18	5	11.64	1.64	5.75	25.20
20	3	11.47	178.00	10.20	21.20
20	4	10.05	114.00	7.79	19.60
20	5	9.67	27.20	6.33	14.30

with (25) removed, i.e., the effect of the predicted RUL of the aircraft engine on the maintenance schedule is not considered. According to Table X, it can be seen that CPLEX can solve the model accurately and give the best maintenance plan in a short time. In addition, different maintenance centers have an impact on the maintenance completion time and maintenance plan. Especially for the MILP-1 model, the maintenance completion time is reduced by $4.85 - 3.46 = 1.39$ when the number of maintenance centers is 3 than 2. Comparing the MILP and MILP-1 models, the required maintenance completion time is then higher when considering (25), but it can more fully utilize the aircraft engine RUL. In other words, the MILP model can fully utilize the aircraft engine RUL before the maintenance alarm threshold is triggered, while the MILP-1 model does not take into account the impact of the predicted RUL of the aircraft engine. Furthermore, comparing the two models, the maintenance scheduling plans obtained are completely different when the maintenance center is 3. For example, the maintenance scheduling plans for the MILP model are 2–6, 1–4, and 3–5, while the maintenance scheduling schemes for the MILP-1 model are 3–5, 2–6, and 1–4.

To validate the effectiveness of the predictive maintenance model, we further test the aircraft engines 11, 18, and 20, which are shown in Table XI. Similarly, the maintenance time of the aircraft engines at the maintenance center is randomly generated within [1.00, 2.00] and the arrival time at the maintenance center is randomly generated within [0.50, 1.00].

From Table XI, CPLEX can accurately solve maintenance models for up to 20 aircraft engines and obtain the best maintenance plan in a short period. In addition, the number of maintenance centers has a different impact on the two predictive maintenance models. In the MILP model, when the number of maintenance centers is increased, the reduction in maintenance completion time is slow due to the influence of the predicted RUL of the last aircraft engine. When the influence of (25) is not taken into account, i.e.,

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in the MILP-1 model, the more the number of maintenance centers is, then the maintenance completion time can be greatly reduced. In conclusion, based on the proposed model, decision-makers can obtain the best maintenance plan for aircraft engines in a short period. Meanwhile, the MILP model or MILP-1 model can be selected according to the needs of decision-makers.

Our proposed predictive maintenance model can be applied to the problem of predictive maintenance of aircraft engines in the aircraft fleet of civil aviation companies and the military aircraft fleet. Meanwhile, the operation status of aircraft engines can be monitored in real-time, so that maintenance can be performed in advance on aircraft engines that are about to break down to avoid causing aircraft accidents. In addition, the proposed predictive maintenance model can simultaneously maintain the aircraft engine in the shortest time.

V. Conclusion

In this article, we propose a complete data-driven predictive maintenance scheduling framework for aircraft engines based on RUL prediction. First, the RUL of the aircraft engine is predicted by BO-CNN-Bi-LSTM-AM ensemble model. Once the aircraft engine predicted RUL triggers the maintenance alarm threshold, the aircraft engine maintenance scheduling task will be immediately initiated. Meanwhile, maintenance priority constraints are proposed based on the size of the aircraft engine RUL, and a new MILP model is developed to deal with the aircraft engine maintenance scheduling problem. Finally, we illustrate the data-driven predictive maintenance scheduling framework for 20 aircraft engines based on RUL prediction. Experimental results show that the proposed data-driven predictive maintenance scheduling framework enables real time aircraft maintenance and reduces maintenance completion time based on aircraft engine RUL prediction and maintenance alarm thresholds, thus fully utilizing the aircraft and preventing failures.

Although this study proposes a novel data-driven predictive maintenance scheduling framework and a MILP model to solve the maintenance scheduling problem for aircraft engines, there are still some shortcomings for further research. In our future research, we will study predictive maintenance scheduling for aircraft engines in more depth. For example, we will consider the impact of aircraft engine mission cycles on predictive maintenance scheduling. Second, the proposed single-objective optimization model can be further extended to a multiobjective optimization model. For example, the maintenance completion time and maintenance center location problem can be considered simultaneously. Finally, a predictive maintenance scheduling decision support system for aircraft engines can also be developed to address the complex operating environment faced by aircraft engines.

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